Histology
of
Cerebellar cortex.

Learning Objectives.

- Mention short review about gross anatomy of cerebellum
- Discuss the layers of cerebellar cortex
- Discuss the cells of cerebral cortex
- Describe Deep Nuclei of Cerebellum

Gross Features of Cerebellum

- Cerebellum is part of brain stem, located within the posterior cranial fossa below the tentorium cerebelli.
- Cerebellum consists of :
- 2 Cerebellar Hemispheres
- Join together by Vermis
- Received input and output through 3 Cerebellar Peduncles which connect three parts of brain stems

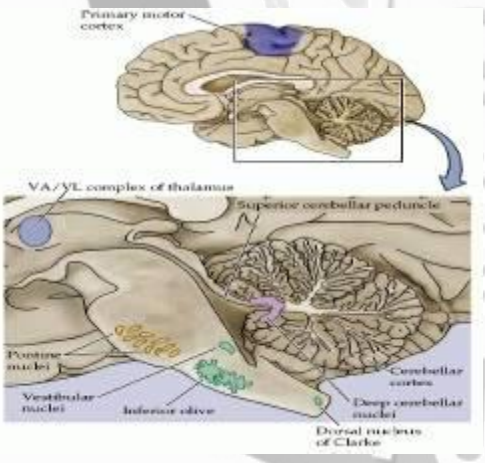
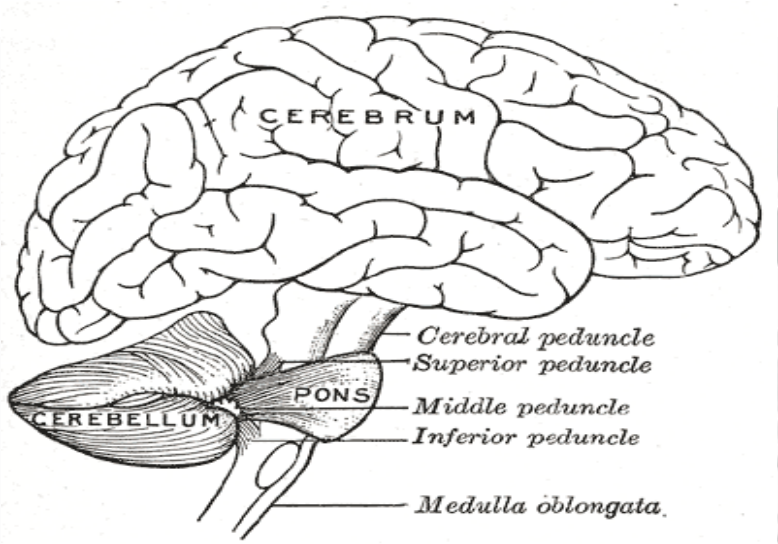


Diagram showing cerebellum.



Histology of Cerebellar Cortex.

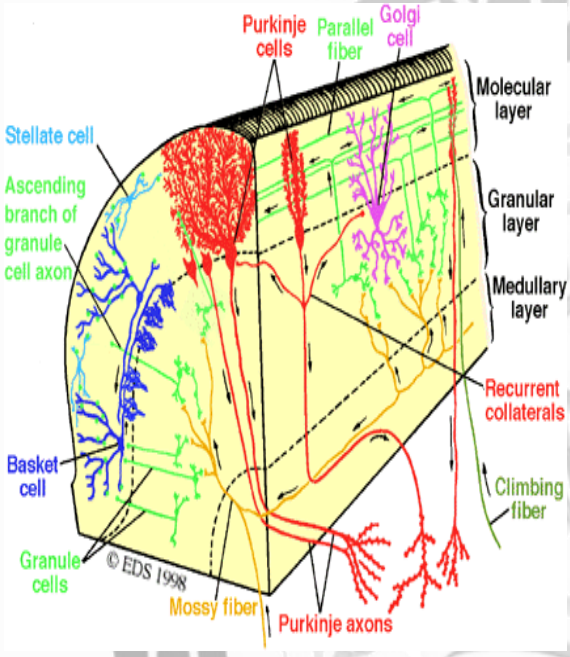
- Surface of cerebellum show transverse fissures which divide the cerebellum into a number of lobules.
- Gray matter of cerebellum is located on the surface as a thin layer called as cortex.
- while the white matter is centrally located, known as medulla.
- There are small masses of gray matter in the centre of white matter called as deep nuclei



Cerebellar Cortex.

- Section of cerebellar cortex shows three layers, from out to inside are:

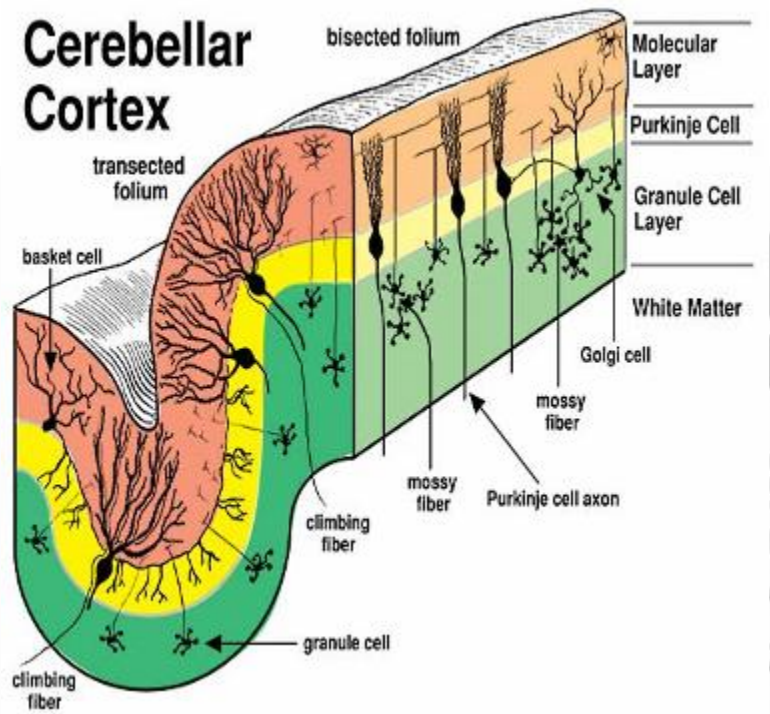
- Molecular Layer.
- Purkinje Cell layer.
- Granular Layer .



1.Molecular Layer

- Outer most layer mainly consists of cell processes.
- These cell process may be dendrites or unmyelinated axon. In this layer make the profuse synapses.
- This layer also contains the two types of cells which are very scanty.

➢ Stellate Cells.
➢ Basket Cells.

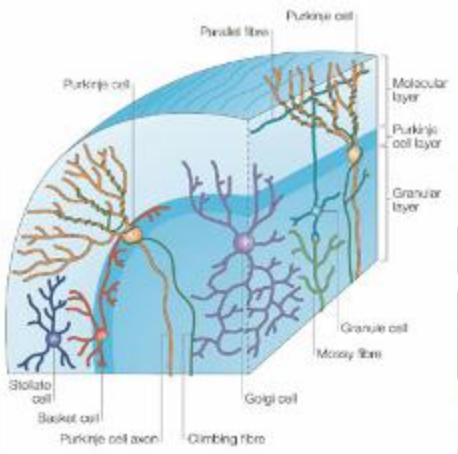


(a)Stellate Cells.

- Stellate cell has small star shaped body with many short dendrites and fine unmyelinated axon.
- The dendrites ramify near the body while the axon extends upward and runs transversely to make synapses with the dendrites of Purknje Cells

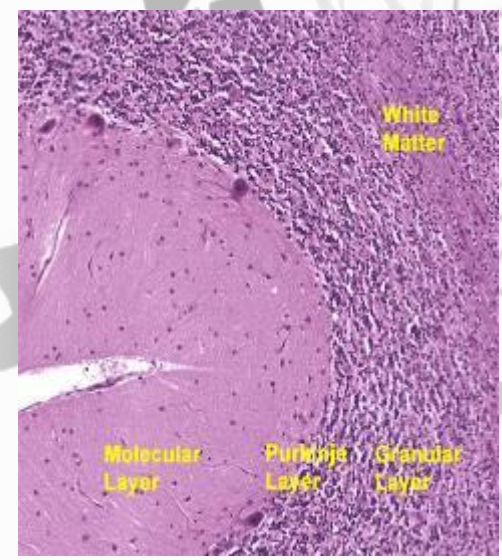
(b) Basket Cells

- They also have small body with numerous branching dendrites that ascend upward toward surface.
- The unmyelinated axon of a basket cell runs horizontally, along it course it gives off many descending branches which form basket like terminal arborizations around the bodies of purkinje cells



2. Purkinje Cell Layer.

- This layer contains the cell bodies of large multipolar neurons called Purkinje cells.
- The purkinje cell consists large flask shaped body.
- The cells are arranged in a single row.
- Each Purkinje cell contains a vesicular nucleus with a prominent nucleus.
- The cytoplasm contains large number of Nissl granules.



2. Purkinje Cell Layer.

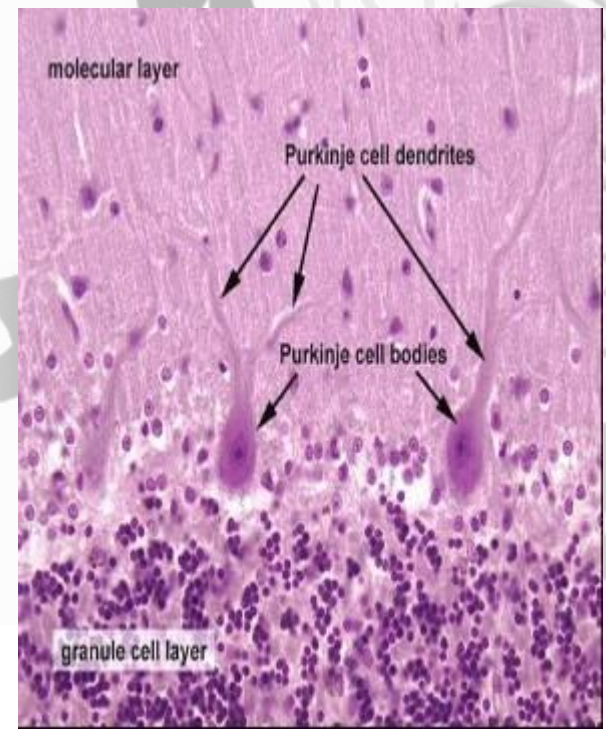
- The dendrites are arising from arise from apex of neuron.
- As two or three large primary dendrites which enter the molecular layer and branched rapidly to form fan-shaped arborization.
- Axon arises from the base of cell, acquires myelin sheath, and passes through the granular layer to enter the underlying white matter.
- Most of the axon of Purkine cells terminate in the deep cerebellar nuclei .

3. Granular Layer.

- This layer consists of thickly populated small cells called granule cells.
- In routine H & E stain the cyoplasm of granule cells cannot be seen and the granular layer appear to be composed of closely packed nuclei which stain deeply basophilic.
- This layer also shows irregular scattered lighter staining area called as glomeruli or island.

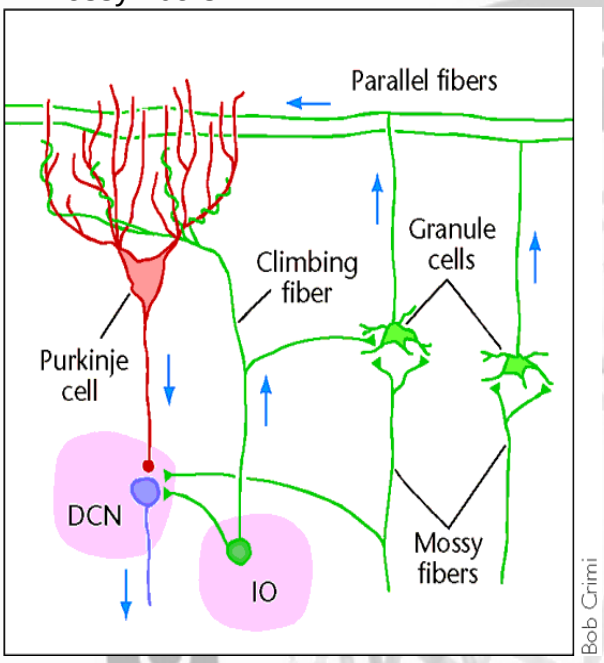
3. Granular Layer(continued).

- Each granular cell gives rise to four or five short dendrites which branch within the glomeruli.
- The unmyelinated axon of granule cell ascends to molecular layer, where it bifurcate into two branches.
- These branches run parallel to the cortex and are known as parallel fibers, make synapses with dendrites of Purkinje cells.



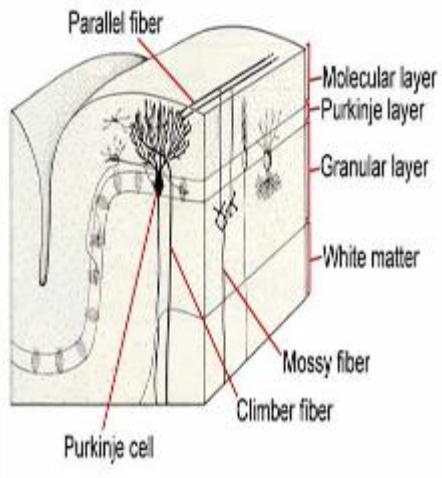
Golgi type II cells.

- Granular layer also contains Golgi type II cells.
- Dendrites of which extend through all layers of the cerebellar cortex.
- The axon of cells end by making synapses with terminal branches of the mossy fibers.



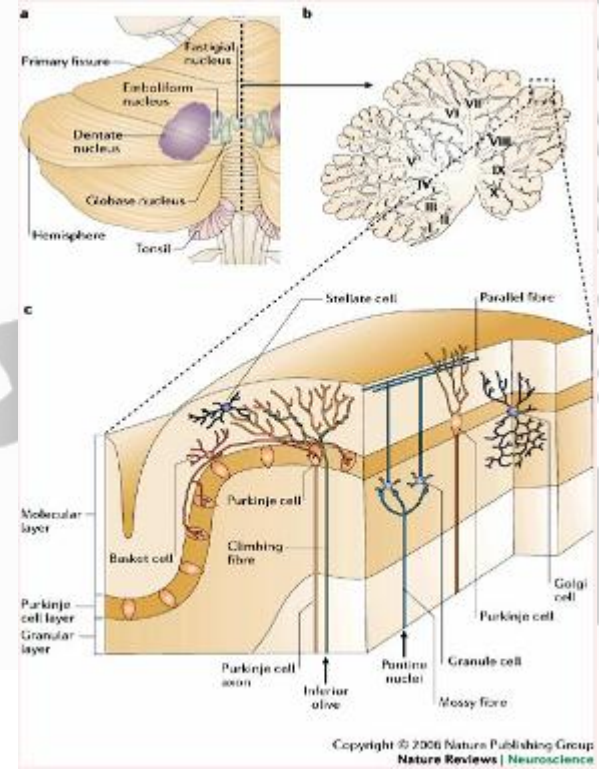
Climbing Fibers.

- Fibers originate in inferior olivary nucleus of the medulla oblongata.
- Passing through the Inferior cerebellar pudencle and enter the cerebellum.
- Then fibers run in the white matter to reach the cerebellar cortex .
- Where they synapse with the dendrites of the Purkinje cells in the molecular layer.



Mossy Fibers.

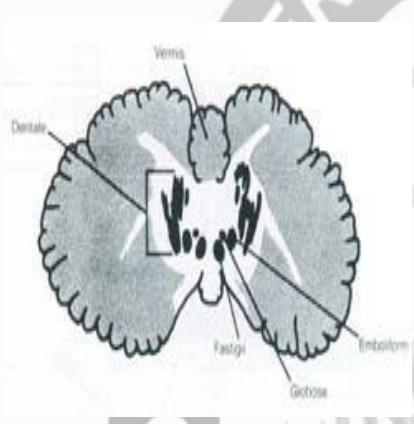
- These fibers originate in the spinal cord (spinocerebellar tract, or in the brain stem (Vestibulocerebellar tract and pontocerebellar fibers
- They terminate in the glomeruli of the cerebellar cortex where they make synaptic contact with the dendrites of the granule cells and axon of Golgi type II cells.



Deep Nuclei of Cerebellum.

- Dentate.
- Emboliform.
- Globose.
- Fastigial.

- **Mnemonic:**
- ‘Don’t Eat Greasy Food’



Self Assesement.

- What connects the Two cerebellar hemispheres?
- What are the three layers of cerebellar cortex?
- Name the cells of molecular layer?
- Name the deep nuclei of cerebellum?